

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis. Synchrony bounds content disagreement only through encoders shared on overlaps, and then only inside a patch small enough for the bound to carry information at all: shrinking a patch tightens the bound and thins its overlaps at the same rate, so the estimate is vacuous across the distant territories whose agreement unity is about. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling, at opposite ends of that observable. A unique fixed point is not a self-model, since a contraction can discard its input: accuracy across a family of states requires distinguishable codes. Where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. And agreement about a change takes communication. Four of the five are checkable

for a machine candidate, which fixes what a claim of machine unity or self-representation commits its maker to measuring. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$; κ is derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs, and an audited axiom footprint no wider than its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and coherent adaptation can lose the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test; the finite-size estimate here is biased below $1/2$ by its escape criterion, as a deterministic reference confirms. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

10 We distinguish three requirements. *Coherence* concerns collective phase
11 order. *Compatibility* concerns whether local descriptions agree where their
12 domains overlap. *Self-representation* concerns whether the resulting state
13 carries an accurate internal reconstruction of itself. We propose the latter
14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 10.3 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these

33 requirements and the assumptions each connection needs. A declared ac-
 34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
 35 overlaps convert phase order into a bound on content disagreement. Compat-
 36 ible spatial measures have a unique global extension; approximate agreement
 37 controls a specified global selection. A contracting reconstruction makes that
 38 extension its unique fixed point only if it accurately reconstructs the exten-
 39 sion. Accuracy across a family of states imposes a further condition on the
 40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
 42 Stationary phase distributions can be shared by stronger coupling, weaker
 43 diffusion and a matched common drive. Finite ramps change the observable
 44 onset. A winding field is stationary at the same coupling as a uniform one and
 45 has the opposite resultant, so that observable does not determine the state.
 46 Coherent adaptation can lose alignment with a designated environmental
 47 template, and a compatibility estimator needs diagnostics that distinguish
 48 content from decoder artifacts. These findings determine the measurements
 49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
 51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
 52 of compatible local sections, and the Banach fixed-point theorem (Banach,
 53 1922). Four things here are not. The installed-energy condition makes cou-
 54 pling a purchased quantity: a declared actuator’s stored energy bounds the
 55 mean coupling it supplies, so coherence on the stationary branch carries a

56 resource requirement in units a measurement can reach, and for cortex that
 57 requirement reduces to a figure the mode prices must deliver. The coherence-
 58 to-content bound converts phase order into a statement about descriptions
 59 rather than about timing, and its patch-local form is what keeps it from
 60 being vacuous in a large population. A stability criterion on the reweighted
 61 coupling kernel, together with an existence condition on an amplitude layer,
 62 settles when a low resultant reports weak coupling and when it reports a
 63 winding field. And the composition is machine-checked together with a wit-
 64 ness discharging all eight hypotheses at once, which is what makes their
 65 conjunction non-empty rather than a list no object satisfies.

66 Granting those premises, collected in Table 1, the account runs as follows.
 67 Extracellular contraction on awakening raises endogenous field amplitude
 68 and with it the effective coupling between cortical oscillators. Once that
 69 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
 70 within a quarter turn the population settles onto it; other initial data leave
 71 windings stationary at the same coupling. Encoders shared where cortical
 72 territories overlap turn that phase order into agreement about the quantities
 73 the territories describe in common, so the local descriptions glue: there is
 74 one state of which each is a restriction. Part of that state encodes the state
 75 itself, and a readout of that part reconstructs it accurately enough, and
 76 sensitively enough across the states that matter, to make it the unique state
 77 the system models itself as being. On this account the glued state is what
 78 makes distributed activity one situation, and its accurate self-reconstruction

79 is what gives that situation a perspective. Every step of it is conditional.

80 Section 2 collects the results that hold whatever physical field is proposed
81 and whatever substrate carries it; a reader who rejects the cortical hypothesis
82 of Section 9 still has to accommodate them. Sections 3–5 then develop the
83 three requirements together (Fig. 1), and Section 6 reads the unconditional
84 results against a machine candidate. Section 7 composes them, stating the
85 thermodynamic constraints and all eight connecting hypotheses per branch;
86 its supporting models of feedback, learning and resource supply have distinct
87 state spaces and assumptions. Section 8 then examines what the observables
88 can establish, before Section 9 develops the cortical hypothesis and an awak-
89 ening experiment. The hypothesis is that extracellular contraction increases
90 field-mediated coupling; its direction, rate and relation to recovery require
91 independent calibration. Supplemental Material organizes the constructions
92 and controls behind the claims they support. Table 2, at the end, lists the
93 symbols and the section that defines each.

94 **2 What holds without the cortical hypothesis**

95 Most of what follows is conditional. Table 1 names eight hypotheses that
96 the composition consumes rather than derives, and the cortical identification
97 of Section 9 is a further commitment on top of them. Five results are not
98 conditional in that way. They constrain any account that reads unity off
99 phase coherence, whatever physical field is proposed to carry the coupling

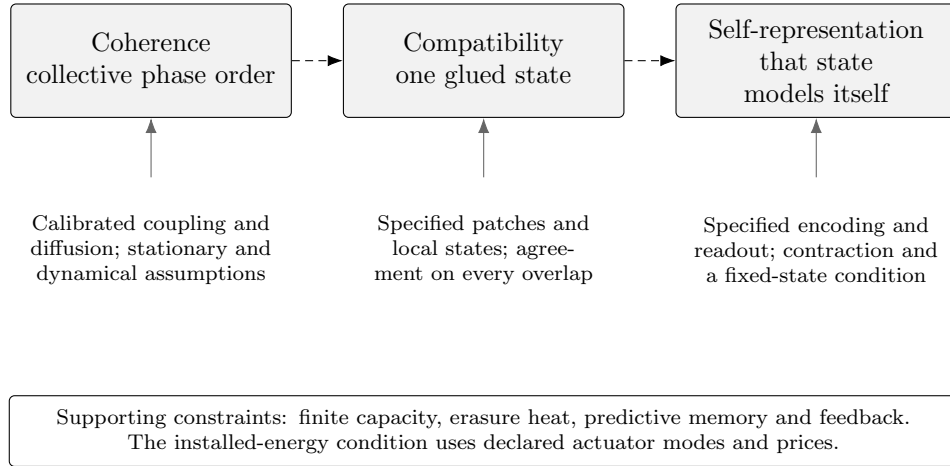


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

100 and whatever substrate carries the field. Each is stated below where it is
 101 used; collected here, they are the article’s fixed points.

102 **Phase order constrains content only inside a patch too small to**
 103 **carry unity.** Assign every patch the same phase and let two local mass
 104 profiles disagree at a shared site. The profiles are chosen independently of
 105 phase, so synchrony places no constraint on their disagreement, and they have
 106 no common extension: perfect locking is consistent with local descriptions
 107 that do not glue (Section 4). What the positive result of Section 4.2 costs is
 108 exactly this gap. A bound on content disagreement needs encoders already

109 related on overlaps, and that relation is a supplied fact about the descriptions
 110 rather than a consequence of the dynamics. Supplying it relocates the limit
 111 rather than lifting it. Eq. (6) bounds a chord, which never exceeds 2, so at
 112 the patch-local order of the sheet in Section 8.3 the bound carries information
 113 only at 3 sites or fewer; shrinking a patch tightens it and thins its overlaps
 114 at the same rate, so the informative end and the end that spans separated
 115 territories are not reached by one cover. At a columnar grain it constrains
 116 agreement inside one locally synchronous population and says nothing about
 117 territories too far apart to share a patch, which is where the compatibility
 118 claim is made.

119 **The order parameter does not determine the state.** A field wind-
 120 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 121 of Eq. (1) below — while every patch of it stays locked, and this is not a
 122 contrived initial condition. On a periodic array of any dimension, a winding
 123 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 124 cause the coupling term is odd in the separation against a kernel even in it
 125 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 126 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 127 diffusion and kernel. A low resultant therefore reports either weak coupling
 128 or a winding field, and no measurement in r alone separates them. The state
 129 also carries an integer the resultant does not: read around a ring, the same
 130 winding has winding number q , and a nonzero q obstructs every global real-

131 valued phase lift (`winding_degree_obstructs`). The resultant is an average
132 over sites and moves when they are reweighted or the patches redrawn; the
133 degree telescopes and does not.

134 **A unique fixed point is not a self-model.** A constant readout con-
135 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
136 is consequently the wrong criterion, and Section 5.2 gives the right shape
137 of one: reconstructing M states pairwise more than 2ε apart within error ε
138 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
139 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
140 readout whatever.

141 **Where the encoding is a restriction, the region must resolve the**
142 **family.** Read the code as the state's own restriction to a region and
143 the criterion moves off the readout entirely. A boundary whose region
144 reads two relevant states identically cannot reconstruct them more than 2ε
145 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
146 boundary that does reconstruct a separated family restricts injectively on
147 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
148 region sits is fixed before an encoding is chosen, so no readout supplies
149 resolution the region lacks.

150 **Agreement after a change takes communication.** In a synchronous
151 network whose updates read only the site's own state and its declared neigh-

152 bours' messages, two configurations agreeing on a site's causal past through
 153 T rounds leave that site identical at round T . Two interventions outside
 154 that past that require reports more than 2ε apart therefore cannot both be
 155 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
 156 ity and timely compatibility are therefore different requirements, and cutting
 157 the path removes the second while leaving the first intact.

158 None of these depends on the coupling being electromagnetic, on the sub-
 159 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
 160 applies four of them — all but the one about the resultant, which is a state-
 161 ment about a spatial observable — to a candidate that is neither biological
 162 nor field-coupled.

163 **3 Coherence: physical coupling and its re-** 164 **source condition**

165 **3.1 Dynamics and the stationary threshold**

166 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

167 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
 168 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,

169 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
 170 quency heterogeneity and diffusion are distinct parameters. The stationary
 171 calculation below uses identical frequencies and this scalar coupling. Ap-
 172 plying it to a spatial kernel requires an aggregation rule and a check of the
 173 resulting approximation.

174 For identical frequencies in their rotating frame, the positive normalized
 175 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

176 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 177 mal derivation uses the absence of stationary current for a periodic gradient
 178 drift to obtain this density, then verifies normalization and self-consistency.
 179 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 180 threshold, exactly one positive solution coexists with zero and increases
 181 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

182 Figure 2 shows this stationary branch.

183 Linearization about the uniform density gives rate $K/2 - D$ for the first
 184 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 185 exactly above threshold. About the coherent density, the quadratic form
 186 is strictly dissipative transverse to collective rotation, whose rate is zero.

187 These linear estimates concern the limiting equation. Relating them to finite
 188 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 189 tion; exchange symmetry alone does not establish the needed independence
 190 (Sznitman, 1991).

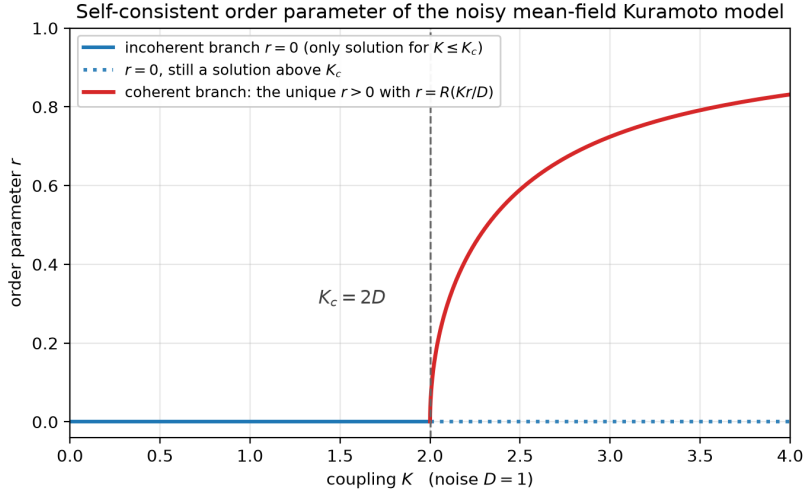


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

191 3.2 Installed energy bounds mean coupling

192 A spatial model replaces the finite sum by integration against a kernel. One
 193 declared implementation carries response modes with continuous profiles ϕ_i ,

194 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

195 The same configuration sets the reciprocal kernel and stored energy. An
 196 executed transition changes the kernel through its changed occupancies and
 197 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 198 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 199 pair reconstruction on refining spatial partitions converges to its product-
 200 measure integral under the specified compactness and sampling assumptions;
 201 the refinement index is independent of the number of control updates.

202 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 203 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 204 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

205 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 206 coherent stationary branch and any growing Fourier mode about the uniform
 207 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 208 is necessary: energy above it need not produce sufficient coupling because
 209 the bound can be loose.

210 The quantity constrained is installed energy rather than dissipated heat,
 211 and the condition binds through κ rather than through the energy, since
 212 any macroscopic substrate carries installed energy far above the thermal
 213 floor. Read forwards it therefore excludes no macroscopic candidate. Read
 214 backwards it is a number. Take the region the field arithmetic takes, a
 215 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
 216 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
 217 at $D = 0.5$ rad²/s: the installed energy is 1.0×10^{-6} J, and the condition is
 218 met at every κ above 9.6×10^5 J⁻¹s⁻¹. That figure is what an identification
 219 of cortical mode prices has to deliver, and the first identification delivering
 220 less refutes the coupling-gated account for cortex at this region and interval.
 221 Another interval rescales the figure by its ratio to this one.

222 What the condition discriminates, it discriminates through that factor.
 223 The required κ scales as D^2 and inversely with the region's cell count, so at
 224 fixed mode prices the condition ranks candidate substrates by their installed
 225 energy; in a declared implementation it decides outright, the learning witness
 226 of Section 7 clearing the line after one update while its matched control stays
 227 on the exclusion side at the same conversion factor and diffusion. What is
 228 missing for cortex is a price list rather than a mechanism. A mode is a priced
 229 rank-one contribution to the kernel, cortical field generation has no canonical
 230 decomposition into such modes, and choosing the basis would choose the
 231 answer. Applying the scalar threshold to heterogeneous or signed spatial
 232 dynamics requires a justified reduction, and identifying the installed kernel

233 with a cortical field is a separate physical commitment.

234 4 Compatibility: from coordinated activity 235 to a global description

236 4.1 Why common timing is insufficient

237 For overlapping patches U and V , compatibility means that restricting either
238 local description to $U \cap V$ gives the same object. Descriptions can agree on
239 shared quantities while contributing different information elsewhere. Com-
240 patible sections of the specified sheaf have a unique global extension. On
241 a finite cover, local finite spatial measures agreeing as measures on every
242 overlap also have a unique finite-measure extension whose sheaf image is the
243 glued section. No mass-one normalization is imposed.

244 In the cup example, visual and reaching descriptions must express a
245 shared quantity, such as cup location, in common coordinates. Restriction
246 retains that quantity while dropping details particular to each description.
247 Agreement permits surface appearance and hand configuration to contribute
248 to one situation. The mathematical counterexample assigns the same phase
249 to every patch while two local mass profiles disagree on a shared site. Those
250 profiles have no common extension. Because they are chosen independently
251 of phase, synchrony places no constraint on their disagreement.

252 This distinction limits the biological interpretation. Receptive territories

253 and content domains must be related to spatial patches, and the descriptions
 254 must support the required extension. Matching overlap marginals of de-
 255 coded probability laws need not determine a unique global joint distribution
 256 (Abramsky & Brandenburger, 2011).

257 The cover is itself a choice, and the construction constrains it in one way
 258 only: synchronized patches agree where they overlap. Where they do not
 259 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 260 satisfies that obligation for *every* assignment of local descriptions and phases
 261 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 262 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 263 ship of the class then reports nothing about the system beyond the patches
 264 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 265 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 266 `Overlaps`), and that what they share is a quantity fixed independently of
 267 the agreement it is being used to establish.

268 4.2 When coherence does constrain content

269 A declared relation between phase and content gives a positive result. Let
 270 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 271 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 272 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 273 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 274 sites; their equality on overlaps is a supplied relation between descriptions of

275 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 276 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 277 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

278 Here discrepancy is the norm of the difference, and r and N are read on
 279 whichever population the bound is applied to. That choice is what decides
 280 whether the estimate says anything. A patch carries an order parameter of
 281 its own, and reading the bound there carries that patch's resultant and that
 282 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 283 `coherence`), so a large population observed through small patches is con-
 284 strained by the small count. Read instead on the whole array, the factor N
 285 makes the estimate weak, because it draws a pointwise guarantee from an
 286 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 287 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 288 tion permits disagreement even at perfect locking. A common nonexpansive
 289 linear projection preserves agreement, but agreement of a projection does
 290 not establish agreement of the full vector.

291 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 292 constrains anything only where $\sqrt{2} N\sqrt{1 - r^2}$ falls below that. The winding
 293 sweep of Section 8.3 fixes both ends of that range on one sheet. Read on the
 294 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$

295 and constrains nothing; holding that order fixed, the count alone brings it
 296 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
 297 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 298 patches therefore thins the overlaps at the same rate as it tightens the bound,
 299 and the two ends are not reached by the same cover. At a columnar grain
 300 the estimate constrains agreement inside one locally synchronous population
 301 and is vacuous across the distant territories whose agreement unity is about.

302 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 303 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 304 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 305 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

306 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 307 duce an initial mismatch. Approximate observations leave a nondecaying
 308 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 309 update to the phase process.

310 **4.3 What approximate agreement determines**

311 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 312 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 313 Weights that vanish outside their patches and sum to one at each site select
 314 a global profile by weighted averaging. The selection lies within ε of each

315 patch there, and any two such selections differ by at most ε , independently
 316 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 317 $L\varepsilon$ across selections.

318 At zero tolerance, the selection is the unique exact extension. At every
 319 positive tolerance there are families with no exact extension; selection then
 320 gives an approximation whose dependence on weights is bounded. Supple-
 321 mental Material gives a three-site witness attaining the bound. The mass-
 322 profile construction uses the scalar case of the content dynamics and addi-
 323 tionally requires nonnegativity of the initial profiles and observations. The
 324 general vector result supplies no identification with the spatial-measure sheaf.

325 These constructions concern declared objects: a specified cover, contents
 326 valued in a normed space, and encoders related on overlaps. They select
 327 neither the neural cover nor the decoder, and identifying their contents with
 328 perceptual content, or establishing that the identification bears on experi-
 329 ence, is a further modelling task; Section 8 sets out what a measurement of
 330 it would have to establish.

331 5 Self-representation: a state that models it- 332 self

333 5.1 The reflexive construction

334 Compatibility gives a global state. Reflexivity relates it to an encoding within
335 that state. Equip global sections with a complete metric. If the additional
336 restriction-resonance condition identifies encoding with restriction to a region
337 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

338 The region and readout are fixed across the state space. If F contracts and
339 fixes the section s_* formed from the local descriptions, Banach’s theorem
340 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
341 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
342 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
343 less than one for a relaxation time $\tau > 0$. Constructing a map with that
344 factor remains a modelling obligation.

345 In the cup example, the global state includes the represented scene and the
346 perceiver’s relation to it. A local encoding must retain enough information
347 for the readout to recover that modelled state; cup position alone would be
348 insufficient if hand configuration and viewpoint could not be reconstructed.

349 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. A blind encoder or constant readout fails it for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a

368 boundary reconstructing two relevant states more than 2ε apart while its re-
 369 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 370 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 371 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 372 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 373 cally is fixed by where the region sits, before an encoding is chosen, so the
 374 resolution the family demands is the region's and no readout supplies it. The
 375 implication runs one way: a region that does resolve the family is not thereby
 376 shown to carry a resonant encoding, and E89 supplies no such resolution.

377 Those two demands are jointly satisfiable with nothing folded anywhere.
 378 Take the region to be the whole substrate: restriction is then the identity, so
 379 the boundary is resonant, reconstructs every declarable family exactly, and
 380 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 381 properness condition on the region (`IsProperAvatar`) is therefore a separate
 382 requirement rather than a consequence of resonance and resolution, and what
 383 a fold into a local region is asked for. Its quantitative form replaces the
 384 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 385 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 386 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 387 number of what the region carries (`encard_le_packingNumber_restrict-`
 388 `ToAvatar`). That is the shape a continuous family of states needs, and at
 389 the whole substrate it says nothing, because the region's range is everything.
 390 The gain L is declared and derived here for no physical readout, and at

391 unbounded gain the bound says nothing.

392 **5.3 Agreement after a change takes communication**

393 Snapshot compatibility does not specify how quickly descriptions respond
394 when a shared quantity changes. In a synchronous network whose updates
395 read only the site’s own state and declared neighbours’ messages, two initial
396 configurations agreeing on a site’s causal past through T rounds produce the
397 same state there at round T . The deadline consequence is the one Section 2
398 states, and it follows from the reconstruction bound in Section 5.2: a report
399 from that site is a readout of a code that the two interventions leave identical.

400 A three-site delay line witnesses failure after one round and exact report-
401 ing after two; cutting the path prevents delivery at every round. A report
402 may still happen to be correct about one fixed world. The bound concerns
403 responsiveness across interventions.

404 Throughout this section *prediction* names a reconstruction map, and the
405 relevant family of states, the metric, the tolerance, the communication graph
406 and the deadline are all declared inputs, as are the represented variables and
407 the fidelity counted as accurate. Reading the map as temporal neural pre-
408 diction would require dynamics; reading the resulting code count as physical
409 memory capacity would require a hardware model, which a state-dependent
410 readout would have to extend.

6 What the conditions constrain in a digital candidate

The results of Section 2 hold for any candidate, and for a large language model (LLM) running on graphics processing units (GPUs) four of them are checkable rather than merely stateable. Those four are where this framework bears on claims made for such systems. It does not exclude them, and no result here could; what it does is fix what a claim of unity or self-representation commits its maker to measuring.

Codes, not reports. Reconstruction across a family is a counting constraint before it is anything else. If two macrostates the candidate is said to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both within ε , whatever the readout. A system’s reports are readouts. Identical verbal reports across two conditions are therefore consistent with identical internal codes and settle nothing about the states, while the converse claim — that the system represents its own condition across a declared family — asserts that its internal code separates that family, which is a statement about activations under intervention and not about text. Naming the channel makes the count concrete. Where the code is the token sequence a model emits—which is what its next step reads—the alphabet is the deployed vocabulary (`tokenChannel`), and a family reconstructed through it is no larger than the vocabulary size raised to the number of steps (`card`).

432 `le_card_tokens`). Over an extended exchange that number is large and the
 433 bound is slack; at a single step it is one vocabulary’s worth, so a claim that
 434 one emission reports the model’s own condition across a declared family as-
 435 serts that the vocabulary separates that family (`card_le_card_tokens_one`).
 436 A claim resting on activations rather than on text has declared a different
 437 channel, to which the same count applies with that channel’s alphabet. The
 438 measurement follows from the statement: declare the family and the fidelity
 439 first, then test whether the encoding region distinguishes its members.

440 **The region, not the readout.** Where the encoding is read as the state’s
 441 restriction to a region of the machine, the demand falls on the region before
 442 any readout exists. For a decoder-only transformer the causal mask makes
 443 the network an instance of the update model of Section 5: the activation at
 444 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
 445 else, so a region’s causal past is fixed by the architecture rather than chosen
 446 by interpretation. That instance is formal, and what it establishes is sharper
 447 than slow growth: the causal past of position t stays inside the positions
 448 at most t at every depth (`mask_ball_subset_le`), so the deadline bound of
 449 Section 5 fires at every deadline at once rather than at a chosen one, with
 450 its communication graph given rather than assumed (`mask_no_guarantee`).
 451 That is a statement about one forward pass; across steps a model does read
 452 its own prior output, which is the channel the preceding paragraph counts.
 453 Read against the coupling proposed here the asymmetry runs both ways, and

454 neither way is an advantage. A kernel of full support has no neighbourhood
 455 structure for the bound to bite on—after one round the causal past is the
 456 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable at
 457 any deadline but zero (`not_outside_past_of_isFullSupport`)—but a physi-
 458 cal field propagates at a finite speed, so treating one that way is a calibration
 459 against the phase time scale and not a theorem. What the comparison does
 460 fix is where the bound has force. Not on the attention graph, where a position
 461 reads every position at or before it and the fan-in is the context length. On
 462 the physical interconnect, where fan-in is genuinely bounded: a graph whose
 463 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 464 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 465 pays a number of rounds growing like $\log_\nu$ of the device count for guaranteed
 466 agreement, and buying that deadline back costs wiring—reaching every site
 467 at a single round requires a device hearing from all but one of the others
 468 (`le_degree_of_reaches_one`). Both ν and the device count are declared, as
 469 the communication graph and the deadline already are, and what the count
 470 bounds is the graph a guarantee requires rather than what hardware is build-
 471 able. A claim that a model reports on its own processing consequently carries
 472 an obligation to say which positions and layers the report’s causal past con-
 473 tains, and a report that appears to track an intervention outside it indicates
 474 a confound or a different region, not a faster channel.

475 **Agreement about what.** Unity requires the candidate’s local descriptions
 476 to agree on independently identified shared quantities (Section 4). Common
 477 timing, a shared residual stream and fluent output supply none of that, for
 478 the reason Section 2 gives: coordination bounds content disagreement only
 479 through encoders already related on overlaps, and which quantities two parts
 480 of a model describe in common is a decoding question carrying the failure
 481 modes of Section 8. The cover is a choice, and choosing it to secure agreement
 482 empties the claim.

483 **What is not excluded.** Digital representations can satisfy compatibility,
 484 so finite hardware is compatible with the gluing construction and none of
 485 the above is a no-go for machines. The coupling-gated route E56–E78 is a
 486 different matter: it requires a calibrated physical coupling, its noise regime
 487 and a connection from coordination to content agreement, and an attention
 488 matrix or a shared clock supplies none of these identifications. Computing a
 489 field model likewise leaves its realization in the computing substrate to be es-
 490 tablished. Those physical obligations fall on the cortical proposal identically.
 491 The capacity comparison those support comparisons lack is available, and it
 492 favours neither side by construction. A continuum code space read at resolu-
 493 tion δ holds at least its volume divided by the volume of a resolution cell, so
 494 it out-resolves an alphabet of 2^b codes exactly when that volume exceeds 2^b
 495 cells (`two_pow_lt_packingNumber_of_lt_measure`); below that threshold the
 496 inequality runs the other way, and at a coarse enough resolution the region

497 holds a single code. What decides the comparison is the volume, the res-
498 olution and the dimension, all measured quantities, and continuity decides
499 nothing on its own: a continuum code space read at a finite resolution is
500 a finite alphabet whose capacity is a bit count. For the same reason this
501 account credits an analog medium with no computation beyond a Turing
502 machine's. Real-valued states carry unbounded capacity only at unbounded
503 precision, and the noise floor $D > 0$ that Eq. (1) is stated against removes
504 it.

505 The installed-energy condition of Section 3.2 is restricted in the same
506 way. It constrains a macroscopic candidate only through κ , which is hard-
507 ware data supplied by mode profiles and prices, and Supplemental Material
508 reads the condition against a cortical metabolic budget to return the conver-
509 sion factor it would require rather than a verdict. The hardware comparisons
510 there have similarly restricted scope: fixed wiring can impose a correlation
511 disadvantage, and point-supported kernels vanish under an atomless contin-
512 uum measure, but neither comparison makes the representation conditions
513 unattainable on GPUs. A universal exclusion would need a necessary condi-
514 tion for consciousness and a proof that every relevant implementation violates
515 it, and the present composition supplies neither.

516 Whether experience requires a living substrate is the explicit subject of
517 biological naturalism (Seth, 2026a,b). On that view the properties ground-
518 ing experience are those of self-sustaining organisms, and simulating them
519 does not secure them. The conditions stated here name no biological mech-

520 anism, but computation described as such does not satisfy them either: a
521 candidate must exhibit a physically realized coupling in a measured noise
522 regime, agreement about independently identified shared quantities, and an
523 internal reconstruction meeting a declared fidelity. Whether a realization
524 of those conditions that is not alive would be accompanied by experience
525 is the point at issue, and these results do not settle it. A related position
526 locates the difference in the mode of computation rather than in life as such:
527 biological systems compute across scales that do not separate, and along-
528 side discrete operations carry continuous-valued dynamics belonging to their
529 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
530 here is one measurable instance of that requirement, fixed to a physical field
531 in a declared noise regime, and a description of such a field does not satisfy
532 it.

533 The neuroscientific case against *human-like* consciousness in text-only
534 LLMs concerns embodied sensory interaction, thalamocortical organization
535 and biological regulation linking perception and action to survival (Aru et al.,
536 2023). Their absence supports skepticism, conditional on their causal rele-
537 vance to experience. Those mechanisms are not requirements of the abstract
538 gluing and reconstruction definitions, and workspace-like representations re-
539 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
540 investigate. What this framework adds to that investigation is an order for
541 the questions: declare the family of states and the fidelity, fix the encoding
542 region and read its causal past off the architecture, and only then ask what

the system reconstructs. A claim that survives those three steps is a claim about the machine; one that does not survive them was a claim about its output.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them, and it does so from the side rather than along the spine: no edge below derives coherence, compatibility or reconstruction from a heat bound, and the resource condition these edges terminate in is necessary without being restrictive for any macroscopic candidate (Section 3.2). What this half contributes is a ledger in which a resource claim is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

562 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 563 et al., 2012). Zero retained information also satisfies it, so a task and a
 564 learning rule require additional specification. Relating this bound to Lan-
 565 dauer heat requires a common thermal scale and budget, and memory waste
 566 bounded by erased entropy.

567 An acting agent changes the signal it will receive. Specify policy, world,
 568 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

569 The second conditional term vanishes in the passive model. With feedback,
 570 the signed difference can be negative as action creates correlations with the
 571 future. A causal reading requires an environmental state sufficient to describe
 572 the relevant dynamics.

573 For a controlled substep, the active thermodynamic bound instead uses
 574 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 575 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 576 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 577 Ito and Sagawa, 2013). Every forward path of positive probability must have
 578 a positive-probability reverse for this finite identity. A missing reverse gives
 579 infinite divergence. The supplement develops the support conditions and
 580 composes actuation with memory updates on their shared intermediate law.

581 **7.2 Learning and repeated operation need explicit re-** 582 **sources**

583 The finite examples distinguish selecting a feasible policy, relaxing a register
584 toward an encoded reward, and learning an unknown task from observa-
585 tions. The observational learner acquires correlation with an environmental
586 parameter while its action marginal stays uniform. A continuing model exe-
587 cutes actuation, observation-driven learning and reset in sequence, preserving
588 above-chance performance at a positive work cost per cycle.

589 A phase-derived sensor makes the informational distinction concrete: an
590 informative readout and a blind control have the same phase order but dif-
591 ferent learning performance. Attaching the learner’s register to the actuator
592 of Section 3.2 ties reward, installed kernel and work to its own executed law.
593 The learning witness rises above the necessary installed-energy bound while
594 its matched control remains on the exclusion line. These finite constructions,
595 detailed in Supplemental Material, leave the evolving feedback between phase
596 dynamics, installation and sensing to be specified.

597 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
598 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

599 where expected cumulative work stays within the supplied allowance B at
600 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one:

620 strictly positive forward and reverse elementary paths, local detailed balance
 621 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 622 witnesses being two-bit systems with interaction energies of that size. Cortex
 623 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 624 temperature over one region and one phase-diffusion time, and a budget B
 625 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 626 satisfied by any macroscopic candidate and excludes none of them. Reading
 627 the register instead at the scale where its premises are physical, such as a
 628 single channel, relocates the difficulty to E45Active, which would then have to
 629 carry one molecular step’s budget to the convergence of a continuum kernel.
 630 The active branch is an alternative route to the same node rather than a
 631 stronger one, and no result of Section 3.2 passes through it: the installed-
 632 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

648 in one place and machine-checked together, and not that coherence follows
649 from installed energy or self-representation from coherence.

650 Joint witnesses combine a register, boundary, refining mesh and small
651 spatial cover; active witnesses add controlled thermal updates and a priced
652 installation. They establish satisfiability under their stated identifications.
653 Mesh convergence follows from refinement data, and the witnesses establish
654 no common cortical mechanism. Supplemental Material maps the claims to
655 their Lean declarations and records the counterexamples. The development
656 declares no additional axioms beyond its libraries' foundational axioms.

657 8 What the observations establish

658 The composition holds under its premises. Whether observation can estab-
659 lish those premises is a separate question, and the three requirements it
660 connects concern different objects: phase dynamics, local descriptions and
661 internal encodings. Their observables must retain these distinctions. The
662 following controls show which mechanisms remain indistinguishable, which
663 temporal approximations fail, which states one observable cannot separate,
664 and what independent content measurements must establish. Each is a finite
665 numerical regime on a declared model, with full parameters and controls in
666 Supplemental Material, and none of them calibrates a cortical quantity.

667 8.1 Stationary distributions leave the mechanism un- 668 determined

669 Equation (2) gives a concentration–coherence relation with no fitted shape
670 parameter. Estimate r from circular resultant length and a from density
671 shape: inverting R from the measured r would make agreement automatic.
672 Even independently estimated agreement identifies a statistical family shared
673 by different mechanisms.

674 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
675 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
676 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
677 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
678 and diffusion D_0 have that same density. Their stationary distributions co-
679 incide window by window despite different causes.

680 Resolved dynamics supply additional information. In the ideal model with
681 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
682 increment means constrain drift. Separating endogenous coupling from a
683 matched common drive additionally requires a measured drive or controlled
684 perturbation; identifying electromagnetic coupling requires evidence specific
685 to that interaction. Supplemental Material gives the equivalence construction
686 and its numerical check.

687 8.2 Finite ramps change the observable onset

688 The square-root law in Eq. (3) describes stationary solutions. A changing
689 coupling requires time for the phase distribution to approach those solutions.
690 In a deterministic mean-field control that crosses the threshold symmetrically,
691 none of the scanned rates meets the declared tracking tolerance: slowing
692 the crossing does not remove the terminal residual under that initialization.
693 A slow control variable therefore supplies no general guarantee of branch
694 tracking at onset. Supplemental Material gives the residual, rate range and
695 admissible rates away from threshold.

696 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
697 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
698 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
699 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
700 6 uncensored legs supply the delay scaling below; the decade-spaced ones
701 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
702 onset exponents differ from those of the exact stationary branch evaluated
703 on the same grids. At the fastest speed the ensemble mean never reaches the
704 fit window, so its fitted slope does not estimate stationary onset.

705 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
706 makes the apparent exponent depend on ramp speed and fitting window.
707 The specified deviation-based critical ramp speed is not bracketed by this
708 sweep. With fluctuating coupling, the relevant stationary comparison also
709 depends on rate: the mean coupling approximates order in the tested fast

710 regime, while averaging stationary order over coupling works for the tested
711 slow drives confined to one side of threshold. Slow crossing drives can satisfy
712 neither approximation.

713 **The prediction.** The delay scaling is what this section supports as a pre-
714 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
715 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$,
716 which does not contain $1/2$. What that shortfall measures is the escape cri-
717 terion and not the scaling. Applied to a deterministic mean-field run seeded
718 at the fluctuation floor, the same estimator returns 0.425 over the same legs
719 at the same escape criterion, inside the interval on the measurement. That
720 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
721 saturation between the floor it starts from and the level it is stopped at, and
722 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
723 where the answer is known, which is what licenses reading the measurement
724 as the law rather than as a competing exponent.

725 Stated as a law rather than as a number: onset delay scales as the square
726 root of the rate at which coupling crosses threshold. Testing it requires
727 only that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor the
728 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15)
729 — so it is a within-subject design that varies the rate of emergence from
730 anaesthesia alone, with each participant as their own control. Such a design
731 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion

732 meets a size-dependent critical fluctuation floor, so the criterion has to be
 733 referred to each recording's own pre-threshold fluctuation scale rather than
 734 to a shared number. The diffusion D is the remaining input, and Section 8.1
 735 notes that it is estimable from resolved phase traces through the rate $2D$
 736 at which quadratic variation accumulates, which is a different quantity from
 737 the concentration the propofol exercise could not range over.

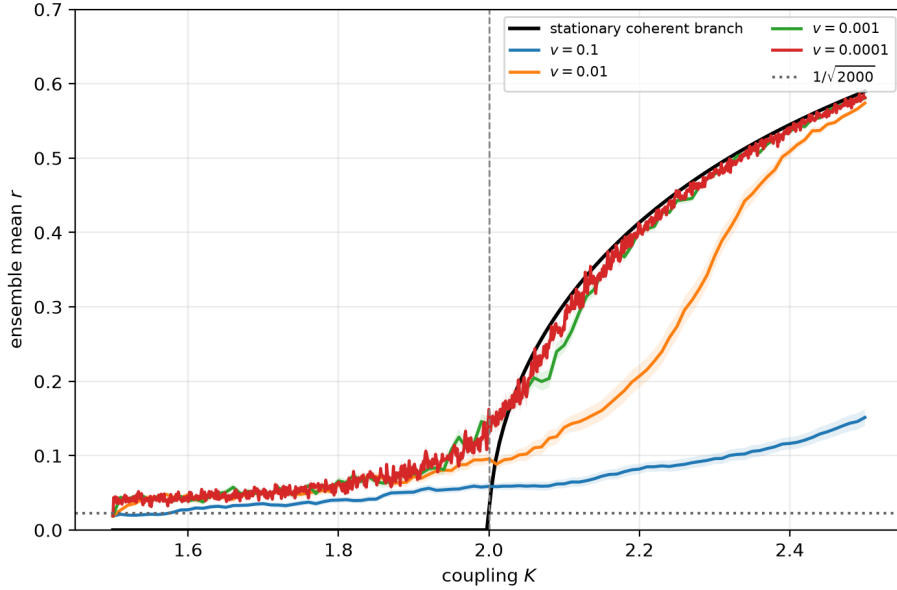


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

738 Additional controls address integration bias, population size and spatial
 739 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 740 $[1.958, 2.009]$ contains 2.

741 8.3 A winding state leaves the order parameter am- 742 biguous

743 The order parameter r is indifferent to where a phase sits in space. A field
744 winding once across a periodic sheet therefore has $r = 0$ while every patch
745 of it stays locked. Such a field is not an artificial construction here. On a
746 torus a winding is stationary for any isotropic symmetric kernel at identical
747 frequencies: the coupling term is odd in the separation against a kernel even
748 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
749 this for any finite periodic array, this sheet included. A winding satisfies the
750 positivity and identical-frequency hypotheses. It violates only the quarter-
751 turn initial spread that the convergence result additionally requires before
752 placing the limit at the uniform state.

753 On the sheet of Section 3's spatial control, a winding survives below
754 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
755 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
756 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
757 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
758 patch-local order 0.9015 and no winding defect. At fixed K the observable
759 therefore takes two values spanning nearly its whole range. Coupling does
760 not determine the resultant, and a low resultant does not by itself distinguish
761 weak coupling from a winding field.

762 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

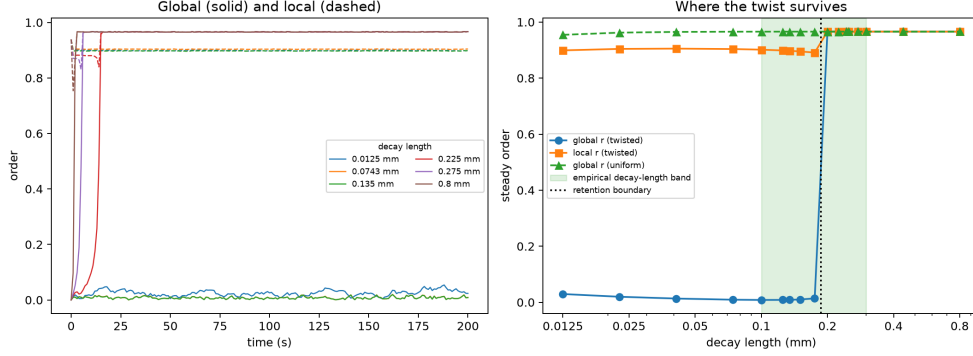


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

763 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 764 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 765 Heterogeneity shortens the range that sustains a wave without removing it.

766 Which windings persist has a criterion behind it. Linearizing the phase
 767 model about a winding returns a coupling of the same translation-invariant
 768 form, each separation reweighted by the cosine of the phase the winding
 769 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 770 again even in the separation, so the same characters diagonalize it and ev-
 771 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 772 at most a quarter turn across every separation the coupling reaches, the

773 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 774 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 775 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 776 longer separations is what carries a phase advance past a quarter turn, and
 777 that is the mechanism the decay-length sweep varies. The criterion bounds
 778 rates rather than trajectories: asymptotic stability is not proved, so the
 779 states that persist occupy a narrower range than it admits, and the bound-
 780 ary quoted above is measured and not derived from it. The same kernel is the
 781 potential's curvature, so a winding at $r = 0$ is not the global minimum the
 782 convergence result places its limit at, and one carrying a growing mode is not
 783 a local minimum either: that result's quarter-turn hypothesis must exclude
 784 a winding rather than merely decline to cover it (`char_not_potential_min`).

785 An amplitude layer decides which of these states exist, and leaves what
 786 the resultant reports unchanged. Every oscillator of the phase model sits on
 787 the unit circle, so a field with an amplitude node has no representation in
 788 those dynamics. The real Stuart–Landau extension supplies one on the same
 789 coupling, and leaves intact the gradient structure the convergence results rest
 790 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 791 phase equation is the Kuramoto equation with each coupling weighted by a
 792 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 793 this one with the amplitude ratios set to one, rather than a separate object.

794 What the added degree of freedom decides is which windings exist at all.
 795 On a periodic array of any dimension, this sheet included, a winding requires

796 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 797 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 798 phase the winding advances across d (`exists_charState_iff`; on the ring
 799 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 800 coupling mass to longer separations lowers it further. A broad enough kernel
 801 therefore removes the winding outright. The sweep above reaches that same
 802 boundary by a different route: the phase model retains every winding at
 803 every coupling, and loses this one under integration instead. The two layers
 804 bound the same regime from opposite sides. A phase recording does not
 805 separate a state the amplitude layer excludes from one the dynamics fails to
 806 sustain.

807 Three limits bound the layer for a reader of the cortical literature. These
 808 states are windings and not spirals, and on the sheet that is a theorem rather
 809 than a shortage of dimensions: equal modulus at every site puts the phase
 810 circulation at exactly zero on every plaquette, and on a periodic sheet the
 811 circulations of *any* field sum to zero, so a lone defect is not a state of this
 812 geometry at all. Relaxational dynamics has no propagation speed, which is
 813 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
 814 form radius, which a measured envelope is not; this work identifies the two
 815 nowhere. Supplemental Material gives the sweep, its uniform control, the
 816 amplitude layer, the standing wave it admits inside the band, the potential's
 817 curvature at a winding and the defect theorems, together with the remaining
 818 limits of both.

819 A vanishing resultant is the absence of global agreement and not of agree-
 820 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
 821 ancy bound of $2\sqrt{2}L|\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
 822 which falls to zero as the same winding is spread over more sites (`chord_le-`
 823 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
 824 Encoders shared on overlaps therefore still bound content disagreement in-
 825 side a patch of a state that no global observable calls coherent. The winding
 826 number itself stays unconnected to content. It is, however, a property of the
 827 state rather than of the cover read on it. On a ring the whole turns dropped
 828 by the winding's affine local lifts are zero at every site but the wrap, where
 829 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 830 every global real-valued phase lift, so one state at one coupling has resultant
 831 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
 832 sites moves the first quantity and redrawing the loop's intermediate transi-
 833 tions leaves the second where it was. The scope is the ring's: this is a loop's
 834 winding and not a spiral, the transitions are those of the affine lifts rather
 835 than of an unwrapped measurement, and the integer is identified with no
 836 cortical observable.

837 This qualifies what a phase trace can establish. An onset criterion stated
 838 in r alone registers a change of basin as readily as a change of coupling.
 839 The protocol of Section 9.3 therefore needs a spatial observable beside the
 840 resultant.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current vanishes while phase-averaged squared drift is positive. Supplemental Material gives the operator, its gradient and the conditions of the descent result.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. Its template is a diagnostic covariance of latent signals, never a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

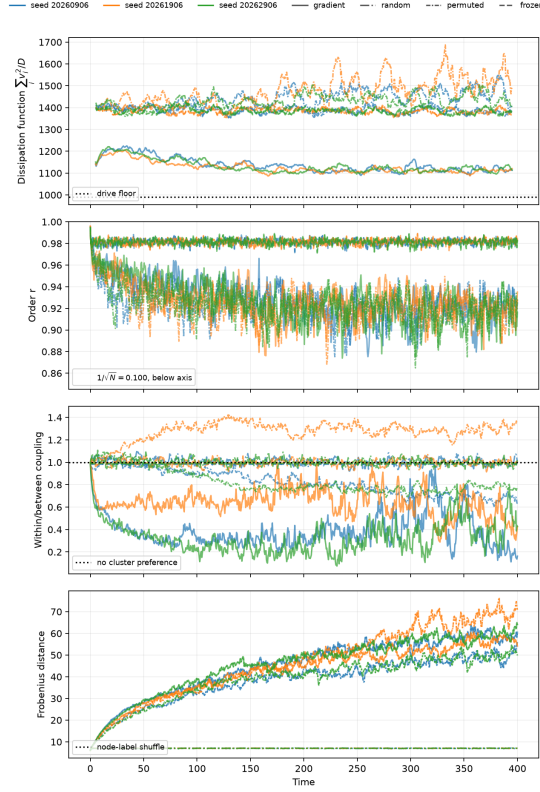


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

885 tent, phase carries a share 1.000 of the content entropy when content is built
886 from it and 0.000 when it is not. A phase-derived statistic reports com-
887 patibility on phase-locked patches whose local descriptions disagree. The
888 implemented content statistic detects their incompatibility. These controls
889 validate a measurement on known constructions; a neural content model and
890 decoder remain to be established.

891 Choosing the cover also chooses a candidate subject: trivial descriptions
892 can agree while carrying little content, and a partition can conceal disagree-
893 ment. The useful choice must therefore be constrained by what neural ac-
894 tivity encodes. The exact composition assumes compatibility of its supplied
895 cover. The content mechanisms above identify additional hypotheses under
896 which agreement can arise; they leave the empirical choice of content and
897 the connection to that cover explicit.

898 **8.6 What an informative phase-density measurement** 899 **requires**

900 The concentration range and the dependence structure decide whether the
901 concentration-coherence relation can be tested at all. Under the specified
902 log-density estimator and bias-floor criterion, discrimination at the observed
903 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
904 designs, rising to 31000 when sites share a mean in groups of 10. These
905 counts describe that estimator and dependence model; the intended recording

906 requires its own calibration. Supplemental Material reports the bin-count
907 controls and the full dependence-aware design.

908 That ceiling comes from exercising the estimators on public propofol-
909 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
910 Bipolar and common-reference montages produce different residuals. The
911 bipolar results are compatible with the Bessel relation over a concentration
912 range too narrow to distinguish it from its linear approximation at the ob-
913 served scatter, which is the range the design scan takes as given. Temporal
914 pooling changes the estimand, and sedation recordings leave the awakening
915 hypothesis untested.

916 An informative dataset therefore needs sufficient concentration range or
917 precision, calibrated estimators and an observation model for dependence,
918 source mixing and reference choice. Public mouse calcium imaging provides
919 thousands of spatially located neurons across waking, sleep and anesthe-
920 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
921 estimation band used here. Site count, state coverage and bandwidth are
922 separate requirements.

923 **9 The cortical hypothesis and an awakening** 924 **test**

925 The cortical proposal identifies a physical source of coupling and asks whether
926 its changes help explain recovery. The tests in Section 8 determine what

927 observations must distinguish before a phase trace can support this identifi-
928 cation.

929 **9.1 Physical identification and calibration**

930 The cortical hypothesis identifies part of effective coupling with endogenous
931 extracellular electromagnetic interactions. Fields can modulate spike timing
932 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
933 propose coordination of distributed neural information (Pinotsis & Miller,
934 2022; McFadden, 2020). Phase organization under propofol motivates study-
935 ing this regime (Bastos et al., 2021), but does not measure coupling.

936 The formal field predicate requires a jointly continuous kernel, a normal-
937 ized substrate, coupling equal to the kernel average, and no single site pro-
938 viding more than half that strength. Toy constructions satisfy it. A cortical
939 realization must specify the scale at which the kernel and its reduction ap-
940 ply: calcium imaging reports different state-dependent modularity patterns
941 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

942 Cortical columns are the obvious candidate unit, and they answer one
943 of the two demands the formalism makes. As the oscillator index they are
944 a serious proposal: they are delineated by response properties and anatomy
945 rather than by the agreement they would be used to establish (Mountcastle,
946 1997), and a phase can be estimated from a column-sized territory at the
947 electrode spacings used for cortical wave measurements (Townsend et al.,
948 2015). As the content cover they are a poor one, for three separate reasons.

949 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
950 the compatibility obligation without constraining anything (Section 4). The
951 cover condition additionally requires the patches to exhaust the substrate,
952 while much of association cortex has no accepted columnar parcellation. And
953 the column's standing as a canonical cortical unit is contested (Horton &
954 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
955 supplies the oscillator index and leaves the content cover to be identified on
956 separate evidence, which is the bridge assumption E78 records.

957 The spatial reduction is also testable. A row-sum aggregation at zero
958 diagonal tracks the uniform-coupling control over the tested spatial decay
959 range, while frequency heterogeneity displaces the operational threshold. The
960 fit window biases both estimates relative to $2D$; agreement with the control
961 is therefore the supported conclusion. Supplemental Material reports the
962 reduction, its numerical error and the distinct assumptions needed for the
963 kernel-average identification in Eq. (5).

964 The field commitment requires physical evidence beyond a choice of math-
965 ematical representation, and the available support comparisons do not supply
966 it. A finite matrix spread over positive-mass cells preserves its total cou-
967 pling and phase correlation exactly, while point-supported vanishing under
968 an atomless measure and a disadvantage from excluding a better-correlated
969 pair concern specified support constraints; Section 6 reads the same compar-
970 isons against a digital candidate. Neither supplies energy prices.

971 A spike-time shift alone supplies no coupling rate. The dimensionless

972 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 973 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

974 This rate describes how induced timing shifts accumulate into drift. Nei-
 975 ther it nor the installation factor κ follows from a heat bound. Testing the
 976 physical threshold requires independent estimates of the interaction kernel,
 977 its aggregation and diffusion in the same cortical regime.

978 **9.2 Extracellular geometry and recovery**

979 The proposed control variable is extracellular geometry. Extracellular vol-
 980 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 981 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 982 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 983 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 984 an effective-medium description predicts that a larger extracellular fraction
 985 raises conductivity and reduces field amplitude.

986 The coupling-gated hypothesis consequently predicts that extracellular
 987 contraction on awakening raises E and, holding the other factors in Eq. (15)
 988 fixed, raises effective coupling. This argument supplies a direction; the
 989 geometry–coupling relation, its magnitude and its role in human recovery
 990 require measurement.

991 For a smooth crossing at nonzero speed, evaluating the stationary branch
 992 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

993 This follows from Eq. (3) at fixed D and concerns the stationary branch
 994 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 995 dently justified tracking approximation. The crossing controls in Section 8.2
 996 leave that approximation unsupported at onset, even though a separation of
 997 geometry and phase time scales can support it away from threshold. The
 998 awakening hypothesis therefore needs a calibrated dynamical prediction for
 999 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 1000 does not supply a recovery latency.

1001 **9.3 A protocol for the awakening experiment**

1002 The physical test has six requirements:

- 1003 **1. Coupling and diffusion in the same units.** Estimate them from re-
 1004 solved phase dynamics, separating heterogeneity and observation noise.
 1005 Measure the drive or perturb the system to distinguish endogenous cou-
 1006 pling.
- 1007 **2. Geometry calibrated to coupling.** Validate the geometry observ-
 1008 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.

4. **An independent recovery endpoint.** Measure behavioural recovery separately from phase organization.

5. **A predeclared dynamical onset criterion.** Set a coherence threshold above the finite-size floor and evaluate its latency under the measured coupling trajectory, including ramp speed or fluctuations. Record a spatial observable beside the resultant, since a winding field reports a low resultant at coupling that sustains local order (Section 8.3).

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A

1030 favorable outcome still leaves the content of coordinated activity to be ex-
1031 plained.

1032 **10 Discussion**

1033 Two things have been established here, and they are of different kinds. The
1034 constraints of Section 2 hold outright: synchrony does not deliver compatible
1035 content, and where shared encoders let it bound content at all it does so only
1036 inside patches too small to span the territories at issue; the resultant does not
1037 determine the state; a unique fixed point is not a reconstruction; a region that
1038 cannot resolve a family cannot encode it; and agreement about a change takes
1039 communication. Any account that reads unity off phase coherence has to
1040 meet them, this one included. The account set out in Section 1 is conditional
1041 at every step by contrast: each connection in it is a hypothesis supplied
1042 rather than a result derived, and Table 1 names all eight. Its four non-
1043 standard ingredients, named in Section 1, are the installed-energy condition,
1044 the patch-local form of the content bound, the criterion separating a winding
1045 field from weak coupling, and the machine-checked composition with its joint
1046 witness. What follows takes the conditions in turn, as commitments the
1047 proposal incurs, as measurements that would bear on them, and as questions
1048 the framework leaves open.

10.1 Unity and a minimal self as empirical commitments

We propose the state glued from compatible local descriptions as a correlate of conscious unity, and its accurate internal reconstruction as a correlate of a minimal self. The motivations are that different contents belong to one experienced situation and that the situation is experienced from a perspective. The conditional results identify relations a realization would need: sufficient installed resources for the specified phase model, agreement about shared content, and an encoding that supports reconstruction at the required fidelity. Discharging E12–E89 for cortex would make the composition a statement about cortex: that a cortical field above threshold relaxes to a cover whose compatible local descriptions glue, and that the glued state is the unique fixed point of a reconstruction map realized there. It would leave the two identifications above exactly where they are, because neither is among the eight and neither follows from their conjunction; they are the commitment this subsection names, and Section 10.2 lists the three observations that would count against it.

Compatibility allows different descriptions to contribute different information. It also permits uncertainty and conflicting judgments: apparent cup position and judged cup position can be distinct quantities in a consistent description of their discrepancy. Which quantities are shared must be determined independently. Choosing them solely to secure agreement would

1071 deprive the proposal of empirical force.

1072 The minimal-self proposal concerns the perceiver’s position and condition
1073 within the situation, such as the cup’s location relative to a body that can
1074 reach it. The encoding could be implicit and need carry no autobiograph-
1075 ical narrative. Its proposed necessity is philosophical: even an unreflective
1076 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1077 constructability alone supplies no functional role. Altering the encoded hand
1078 or viewpoint while holding the external scene fixed should change estimates
1079 of the perceiver’s relation to the cup as the readout predicts. Whether that
1080 represented perspective is experienced remains a further question.

1081 **10.2 What a test of the full proposal must establish**

1082 Both uniqueness results concern specified objects. They choose neither the
1083 cover nor the encoding, and do not select between candidate subjects. In-
1084 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1085 here selection requires independent evidence about neural organization. A
1086 constant readout can contract and fix a predetermined state while discard-
1087 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1088 this construction lacks, provided the relevant states and fidelity are declared
1089 independently.

1090 A test must separately measure coupling relative to diffusion, content
1091 compatibility and self-reconstruction, alongside an independent endpoint for
1092 experience. Meeting all three without a unified episode would challenge suffi-

1093 ciency. A unified episode with incompatible descriptions would challenge the
1094 unity identification; one lacking the specified self-encoding would challenge
1095 the reflexive requirement. Awakening, dreaming and dissociative states offer
1096 possible settings, but absence of report cannot by itself define absence of
1097 experience.

1098 The next physical requirement is a calibrated geometry–coupling relation
1099 and resolved phase dynamics. The installation bound contributes a nec-
1100 essary resource condition where its hardware representation applies; cortical
1101 mode prices and field-mediated coupling require measurement. For the repre-
1102 sentational connection, an informative experiment asks whether a controlled
1103 change in coordination alters independently decoded content agreement, with
1104 controls for sensory information and decoder quality. The constructed-data
1105 estimator establishes feasibility for specified contents. Neural local descrip-
1106 tions, shared overlap encoders and the self-encoding have yet to be identified.
1107 The population-size dependence of the phase-to-content bound further limits
1108 its quantitative use, in the form Section 4.2 reads off the sheet.

1109 The separation results explain why these measurements must be inde-
1110 pendent. A stationary distribution leaves several recovery mechanisms indis-
1111 tinguishable. Equal phase order can accompany incompatible descriptions
1112 or different task information. The tested plasticity rule preserves coherence
1113 while losing its designated template. Each finding concerns the specified
1114 model and observable; their role is to delimit the evidence needed for the
1115 proposed connections.

10.3 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. That permission is wider than the unconditional results of Section 2 alone. The conditional chain is substrate-neutral in the same way: its field condition asks for a jointly continuous kernel on a normalized substrate, a coupling equal to that kernel’s double average, a noise that is the field’s own, and no single site carrying more than half the coupling—and none of these distinguishes an electromagnetic kernel from any other continuous mean-field kernel. The site condition is the only one with physical flavour, and it holds vacuously wherever the substrate has no atoms (`no_site_dominates`), which is the continuum case the electromagnetic hypothesis is about. The chain therefore transfers to extracellular ion diffusion, to gap junctions modelled as a density, or to a mechanical or optical medium, modulo the calibration of Section 9.1. What carries the electromagnetic identification is that calibration together with the extracellular-geometry argument of Section 9, which is the one place a field equation enters. The formalization checks implications under their hypotheses, and its witnesses establish realizability on declared models. Two things stand outside that result and are not supplied by it: the

1139 eight hypotheses are discharged for toy components rather than for cortex,
1140 and the identification of the glued state with unity and of its reconstruction
1141 with a minimal self is a further commitment that the eight do not contain.

1142 The explanatory question is whether the specified relations occur in neural
1143 activity and matter for experience. In the cup example, this means establish-
1144 ing how coordination helps maintain one represented situation and how the
1145 perceiver's place within it contributes to the episode. The proposal identi-
1146 fies candidate correlates and tests of that identification. Why those relations
1147 would be accompanied by experience remains unexplained.

1148 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1)	Phase diffusion, in inverse time
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring

Table 2 (continued)

Symbol	Defined	Meaning
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	Section 5	Number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
E, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification

Table 2 (continued)

Symbol	Defined	Meaning
U	§3.2, §4	Stored energy $U(\Xi)$ of a configuration; a patch U of the cover
s, s_*	§4–5	A global section and the one the reconstruction fixes

1149 Data and code availability

1150 The Lean development, simulation scripts, tests and saved numerical sum-
1151 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1152 `cortical_coherence`. Publication simulation values are generated from
1153 saved summaries without rerunning production sweeps. The exploratory
1154 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1155 processing and selection are described in Supplemental Material. The repos-
1156 itory contains the reproducible analysis and does not redistribute the raw
1157 recordings.

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1159 This research received no external funding. The author declares no compet-
1160 ing interests.

1161 Use of AI tools

1162 AI assistance to this work was substantial and is not separable by task. Sev-
1163 eral coding assistants and chat interfaces were used in an interleaved way
1164 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1165 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1166 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1167 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1168 accessed with the Hermes CLI. Between them they contributed to the Lean
1169 development, the simulation and analysis code, and the drafting and revision
1170 of both documents. Because the tools were used in combination and in alter-
1171 nation, no section, proof, figure or result is attributable to any one of them,
1172 and none is claimed to be. The author is responsible for all content, including
1173 everything a tool produced. Machine checking establishes the stated Lean
1174 results under their hypotheses; it does not validate biological interpretations
1175 or replace scientific review of generated code and prose.

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